Laplace Transform — Role in Control System Analysis

Mini Paper — Power Electronics Fundamentals

Objective

This note summarizes the role of the Laplace transform in control system modeling and analysis.
The focus is on:

- Converting differential equations into algebraic form
- Interpreting system behavior in the s-domain
- Understanding operational properties relevant to control design

The goal is to establish a working analytical foundation for transfer functions and feedback systems.

Definition

For a time-domain function $f(t)$where $t \geq 0$:

$$F(s) = \int_0^{\infty} f(t)e^{-st}\ dt$$

The inverse transform reconstructs the original signal:

$$f(t) = \mathcal{L}^{-1}\{F(s)\}$$

Simplified notation:

$$F(s) = \mathscr{L}\{f(t)\},\ f(t) = \mathscr{L}^{-1}\{F(s)\}$$

Here, $s = \sigma + j\omega$ is the complex frequency that combines exponential and oscillatory components, allowing simultaneous time and frequency representation.

Basic Functions and Their Transforms

Function ($\boldsymbol{f}(t)$)	Laplace Transform ($\boldsymbol{F}(s)$)
$\mathcal{L}\{1\}$	$\frac{1}{s}$
$\mathcal{L}\{t\}$	$\frac{1}{s^2}$
$\mathcal{L}\{e^{-at}\}$	$\frac{1}{s+a}$

$\mathcal{L}\{\cos(\omega t)\}$	$\frac{s}{s^2+\omega^2}$
$\mathcal{L}\{\sin(\omega t)\}$	$\frac{\omega}{s^2+\omega^2}$

Fundamental Signals and Their Meanings

(1) Impulse Function $\delta(t)$

An infinitely narrow, unit-area pulse that models an instantaneous input.

$$\mathcal{L}\{\delta(t)\} = 1$$

It's used to measure a system's instantaneous reaction, forming the foundation for impulse response analysis.

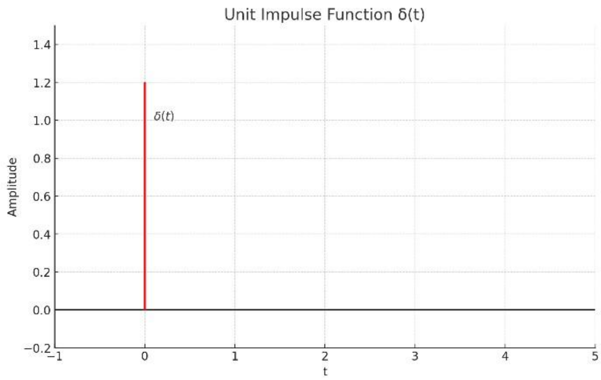

Figure 1 Unit Impulse Function

(2) Step Function $u(t)$

Represents a sudden change, such as switching a voltage source at $t = 0$:

$$\mathcal{L}\{u(t)\} = \frac{1}{s}$$

It is the most common test input in control engineering for checking steady-state performance.

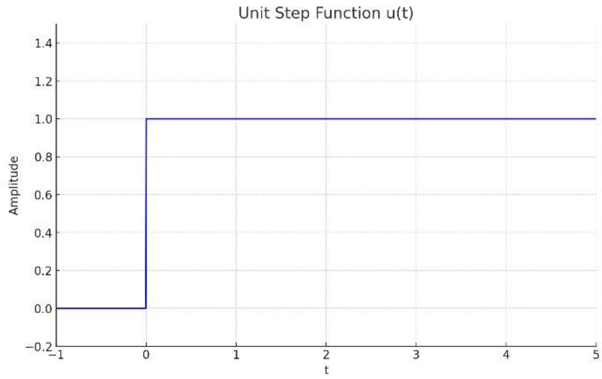

Figure 2 Unit Step Function

(3) Ramp Function $tu(t)$

Represents a linearly increasing input:

$$\mathcal{L}\{t\,u(t)\} = \frac{1}{s^2}$$

Used to evaluate a system's ability to follow gradually changing inputs.

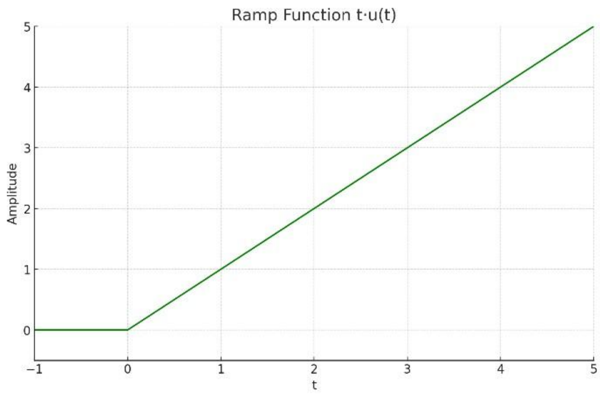

Figure 3 Ramp Function

Higher-order inputs (parabolic, sinusoidal) are used to evaluate dynamic tracking and frequency response.

Core Operational Properties

Linearity

$$\mathscr{L}\{K_1 f_1(t) + K_2 f_2(t)\} = K_1 F_1(s) + K_2 F_2(s)$$

This allows superposition of multiple signals, which is essential for system modeling.

Differentiation and Integration in the Laplace Domain

Differentiation

Differentiation in time corresponds to multiplication by $s$in the Laplace domain:

$$\mathscr{L}\{f'(t)\} = sF(s) - f(0)$$
$$\mathscr{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0)$$

Each derivative introduces the initial condition terms, which makes Laplace analysis particularly powerful for systems with known starting values.

Integration

Integration in time corresponds to division by s:

$$\mathcal{L}\left\{\int_0^t f(\tau)\ d\tau\right\} = \frac{F(s)}{s}$$

Initial and Final Value Theorems

$$f(0^+) = \lim_{s\to\infty} s\,F(s)$$

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} s\,F(s)$$

Used to determine system start-up and final response without full inverse transformation.

Solving Differential Equations

Example:

$$2\frac{dx(t)}{dt} + 3x(t) = 0, x(0) = 1$$

Taking Laplace:

$$2[sX(s) - 1] + 3X(s) = 0$$
$$X(s) = \frac{2}{2s + 3}$$

Inverse transform:

$$x(t) = e^{-1.5t}$$

The differential equation is reduced to algebraic manipulation in the s-domain.

Inverse Transform via Partial Fractions

Any rational function:

$$\boldsymbol{F}(s) = \frac{N(s)}{D(s)}$$

can be decomposed into simple fractions:

$$F(s) = \sum \frac{a_i}{s - p_i}$$

which corresponds in time domain to:

$$f(t) = \sum a_i e^{p_i t}$$

Repeated poles introduce time-multiplied terms such as:

$$te^{-t}$$

This connects pole locations directly to time-domain behavior.

Summary

• The Laplace transform converts differential equations into algebraic expressions.
• The s-variable captures both exponential decay/growth and oscillation.
• Differentiation and integration become simple algebraic operations.
• Initial and final value theorems simplify stability and steady-state analysis.
• The method forms the analytical basis for transfer functions and feedback control.

This provides the mathematical foundation required for transfer-function modeling and loop design.

References

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